

Appendix F

Linear response: examples

F.1 Dissipation

Many experimental techniques probe the dynamics of a many-body system by measuring the absorption of some externally applied field (e.g., visible light, infrared radiation, microwave radiation). Linear response theory allows us to establish a simple relation between the absorption spectrum and the Fourier transform of a time-correlation function. To see this, let us again consider an external field that is coupled to a dynamical variable $A(\mathbf{p}^N, \mathbf{q}^N)$. The time-dependent Hamiltonian, $\mathcal{H}$, of the system is

$$\mathcal{H}(t) = \mathcal{H}_0 - f(t)A(\mathbf{p}^N, \mathbf{q}^N).$$

Note that the only quantity explicitly time-dependent is $f(t)$. As the Hamiltonian depends on time, the total energy E of the system also changes with time:

$$E(t) = \langle \mathcal{H}(t) \rangle.$$

Let us compute the average rate of change of the energy of the system. This is the amount of energy absorbed (or emitted) by the system, which per unit of time reads:

$$\begin{aligned} \frac{\partial E}{\partial t} &= \left\langle \frac{d\mathcal{H}}{dt} \right\rangle \\ &= \left\langle \sum_i \left(\dot{\mathbf{q}}_i \frac{\partial \mathcal{H}}{\partial \mathbf{q}_i} + \dot{\mathbf{p}}_i \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \right) + \frac{\partial \mathcal{H}}{\partial t} \right\rangle. \end{aligned} \tag{F.1.1}$$

But, from Hamilton's equations of motion, we have

$$\dot{\mathbf{q}}_i = \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i}$$

and

$$\dot{\mathbf{p}}_i = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}_i}.$$

As a consequence, Eq. (F.1.1) simplifies to

$$\frac{\partial E}{\partial t} = \left\langle \frac{\partial \mathcal{H}}{\partial t} \right\rangle$$

$$\begin{aligned}
&= -\left\langle \dot{f}(t) A(\mathbf{p}^N, \mathbf{q}^N) \right\rangle \\
&= -\dot{f}(t) \langle A(t) \rangle.
\end{aligned} \tag{F.1.2}$$

Note, however, that $\langle A(t) \rangle$ itself is the response to the applied field f :

$$\langle A(t) \rangle = \int_{-\infty}^{\infty} dt' \chi_{AA}(t-t') f(t').$$

Let us now consider the situation where $f(t)$ is a periodic field with frequency ω (e.g., monochromatic light). In that case, we can write $f(t)$ as

$$f(t) = \text{Re } f_{\omega} e^{i\omega t}$$

and

$$\dot{f}(t) = \frac{i\omega}{2} \left(f_{\omega} e^{i\omega t} - f_{\omega}^* e^{-i\omega t} \right).$$

The average rate of energy dissipation is

$$\begin{aligned}
\frac{\partial E}{\partial t} &= -\dot{f}(t) \langle A(t) \rangle \\
&= -\dot{f}(t) \int_{-\infty}^{\infty} dt' \chi_{AA}(t-t') f(t').
\end{aligned} \tag{F.1.3}$$

For a periodic field, we have

$$\begin{aligned}
\int_{-\infty}^{\infty} dt' \chi_{AA}(t-t') f(t') &= \frac{f_{\omega} e^{i\omega t}}{2} \int_{-\infty}^{\infty} dt' \chi_{AA}(t-t') e^{i\omega(t'-t)} \\
&\quad + \frac{f_{\omega}^* e^{-i\omega t}}{2} \int_{-\infty}^{\infty} dt' \chi_{AA}(t-t') e^{-i\omega(t'-t)} \\
&= \pi \left[f_{\omega} e^{i\omega t} \chi_{AA}(\omega) + f_{\omega}^* e^{-i\omega t} \chi_{AA}(-\omega) \right],
\end{aligned} \tag{F.1.4}$$

where

$$\chi_{AA}(\omega) \equiv \frac{1}{2\pi} \int_0^{\infty} dt \chi_{AA}(t) e^{-i\omega t}.$$

To compute $\dot{E}$, the rate of change of the energy, we must average $\langle \partial \mathcal{H} / \partial t \rangle$ over one period, $T (= 2\pi/\omega)$, of the field:

$$\begin{aligned}
\dot{E} &= \frac{-\pi}{2T} \int_0^T dt \left\{ i\omega (f_{\omega} e^{i\omega t} - f_{\omega}^* e^{-i\omega t}) \right. \\
&\quad \times \left. \left[f_{\omega} e^{i\omega t} \chi_{AA}(\omega) + f_{\omega}^* e^{-i\omega t} \chi_{AA}(-\omega) \right] \right\} \\
&= -\pi \omega |f_{\omega}|^2 \frac{\chi_{AA}(\omega) - \chi_{AA}(-\omega)}{2i}
\end{aligned}$$

$$= -\pi\omega |f_\omega|^2 \text{Im}[\chi_{AA}(\omega)]. \quad (\text{F.1.5})$$

We use the relation between $\chi_{AA}(t)$ and the autocorrelation function (2.5.17) of A :

$$\chi_{AA}(\omega) = \frac{1}{2\pi} \int_0^\infty dt e^{-i\omega t} [-\beta \langle A(0) \dot{A}(t) \rangle].$$

The imaginary part of $\chi_{AA}(\omega)$ is given by

$$\begin{aligned} \text{Im}[\chi_{AA}(\omega)] &= \frac{\beta}{2\pi} \int_0^\infty dt \sin(\omega t) \langle A(0) \dot{A}(t) \rangle \\ &= -\frac{\beta}{4\pi} \int_{-\infty}^\infty dt \omega \cos(\omega t) \langle A(0) A(t) \rangle. \end{aligned} \quad (\text{F.1.6})$$

Finally, we obtain

$$\dot{E} = |f_\omega|^2 \frac{\beta\omega^2}{4} \int_{-\infty}^\infty dt \cos(\omega t) \langle A(0) A(t) \rangle. \quad (\text{F.1.7})$$

Hence, from knowledge of the autocorrelation function of the quantity that couples with the applied field, we can compute the shape of the absorption spectrum. This relation was derived assuming classical dynamics and therefore is valid only if $\hbar\omega \ll k_B T$. However, it is also possible to derive a quantum-mechanical version of linear response theory that is valid for arbitrary frequencies (see, e.g., [53]).

To give a specific example, let us compute the shape of the absorption spectrum of a dilute gas of polar molecules. In that case, the relevant correlation function is the dipole (μ) autocorrelation function:

$$\langle M_x(0) M_x(t) \rangle = \frac{N}{3} \langle \mu(0) \cdot \mu(t) \rangle.$$

For molecules that rotate almost freely (*almost*, otherwise there would be no dissipation), $\mu(0) \cdot \mu(t)$ depends on time, because each molecule rotates. For a molecule with a rotation frequency ω , we have

$$\mu(0) \cdot \mu(t) = \mu^2 \cos(\omega t),$$

and for an assembly of molecules with a thermal distribution of rotational velocities $P(\omega)$, we have

$$\langle \mu(0) \cdot \mu(t) \rangle = \mu^2 \int d\omega P(\omega) \cos(\omega t).$$

The rate of absorption of radiation is then given by

$$\dot{E} = \frac{\pi\beta\omega^2 N \mu^2}{12} P(\omega) |f_\omega|^2. \quad (\text{F.1.8})$$

For more details about the relation between spectroscopic properties and time correlation functions, the reader is referred to the article by Madden in [44].

F.2 Electrical conductivity

In the derivation of linear response theory in Chapter 2, we assumed that we prepare the system in an equilibrium state with the perturbation *on* and then allow the system to relax to a new equilibrium state with the perturbation *off*. However, this will not always work. Consider, for instance, electrical conductivity. In that case, the perturbation is an electrical field that will cause a current to flow in the system. Hence, the state in which we prepared the system with the field on is *not* an equilibrium state but a steady *non-equilibrium* state. The same holds, for instance, for a system under steady shear. It would seem that, in such circumstances, one cannot use the framework of linear response theory in its simplest form to derive transport coefficients such as the electrical conductivity σ_e or the viscosity η . Fortunately, things are not quite as bad as that. Consider, for example, electrical conductivity. Indeed, if we put a conducting system in an external field, we will generate a non-equilibrium steady state. However, what we *can* is perturbing the system by switching on a weak, uniform *vector potential* $\mathbf{A}$. The Hamiltonian of the system with the vector potential switched on is

$$\mathcal{H}' = \sum_{i=1}^N \frac{1}{2m_i} \left(\mathbf{p}_i - \frac{e_i}{c} \mathbf{A} \right)^2 + \mathcal{U}_{\text{pot}}. \quad (\text{F.2.1})$$

The system described by this Hamiltonian satisfies the same equations of motion as the unperturbed system ($\mathbf{A}$ is a gauge field), and the system will be in an equilibrium state at $t = 0$. We then abruptly switch off the vector potential. From electrodynamics, we know that a time-dependent vector potential generates an electric field:

$$\mathbf{E} = -\frac{1}{c} \dot{\mathbf{A}}. \quad (\text{F.2.2})$$

In the present case, the electrical field will be an infinitesimal δ spike at $t = 0$:

$$\mathbf{E}(t) = \frac{1}{c} \mathbf{A} \delta(t). \quad (\text{F.2.3})$$

We can compute the current that results in the standard way. We note that we can write $\mathcal{H}'$ in Eq. (F.2.1) as

$$\begin{aligned} \mathcal{H}' &= \mathcal{H}_0 - \sum_{i=1}^N \frac{e_i}{cm_i} \mathbf{p}_i \cdot \mathbf{A} + \mathcal{O}(A^2) \\ &= \mathcal{H}_0 - \frac{\mathbf{A}}{c} \int d\mathbf{r} \sum_{i=1}^N \frac{e_i}{m_i} \mathbf{p}_i \delta(\mathbf{r}_i - \mathbf{r}) \end{aligned}$$

$$= \mathcal{H}_0 - \frac{\mathbf{A}}{c} \int d\mathbf{r} \mathbf{j}(\mathbf{r}), \quad (\text{F.2.4})$$

where $\mathbf{j}(\mathbf{r})$ denotes the current density at point $\mathbf{r}$. The average current density at time t due to the perturbation is given by

$$\langle \mathbf{j}(t) \rangle = \frac{\mathbf{A}}{c V k_B T} \int d\mathbf{r} d\mathbf{r}' \langle \mathbf{j}(\mathbf{r}, 0) \mathbf{j}(\mathbf{r}', t) \rangle. \quad (\text{F.2.5})$$

The phenomenological expression for the current response to an applied δ -function electric field spike is (see Eq. (2.5.14))

$$\begin{aligned} \langle \mathbf{j}(t) \rangle &= \int_{-\infty}^t dt' \sigma(t-t') E(t') \\ &= \sigma(t) \frac{\mathbf{A}}{c}. \end{aligned} \quad (\text{F.2.6})$$

From this, it immediately follows that

$$\sigma(t) = \frac{1}{V k_B T} \int d\mathbf{r} d\mathbf{r}' \langle \mathbf{j}(\mathbf{r}, 0) \mathbf{j}(\mathbf{r}', t) \rangle. \quad (\text{F.2.7})$$

The dc conductivity is then given by

$$\sigma(\omega = 0) = \frac{1}{V k_B T} \int_0^\infty dt \int d\mathbf{r} d\mathbf{r}' \langle \mathbf{j}(\mathbf{r}, 0) \mathbf{j}(\mathbf{r}', t) \rangle. \quad (\text{F.2.8})$$

F.3 Viscosity

The corresponding linear response expression for the viscosity seems more subtle because shear usually is not interpreted in terms of an external field acting on all molecules. Still, we can use, by analogy to the electrical conductivity case, a canonical transformation, the time derivative that corresponds to uniform shear. To achieve this, we consider a system of N particles with coordinates $\mathbf{r}^N$ and Hamiltonian

$$\mathcal{H}_0 = \sum_{i=1}^N p_i^2 / (2m_i) + \mathcal{U}(\mathbf{r}^N). \quad (\text{F.3.1})$$

Now consider another system described by a set of coordinates $\mathbf{r}'^N$ related to $\mathbf{r}^N$ by a linear transformation:

$$\mathbf{r}'_i = \mathbf{h} \mathbf{r}_i. \quad (\text{F.3.2})$$

The Hamiltonian for the new system can be written as

$$\mathcal{H}_1 = \sum_{i=1}^N \frac{1}{2m_i} \mathbf{p}'_i \cdot \mathbf{G}^{-1} \cdot \mathbf{p}'_i + \mathcal{U}(\mathbf{r}'^N), \quad (\text{F.3.3})$$

where G , the metric tensor, is defined as

$$G \equiv h^T \cdot h. \quad (\text{F.3.4})$$

We assume that h differs infinitesimally from the unit matrix I :

$$h = I + \epsilon. \quad (\text{F.3.5})$$

In the case that we are interested in the effect of uniform shear, for instance, we could choose $\epsilon_{xy} = \epsilon$, while all other elements of $\epsilon_{\alpha\beta}$ are 0. Now consider the case that we equilibrate the system with Hamiltonian $\mathcal{H}_1$, and at time $t = 0$, we switch off the infinitesimal deformation ϵ . This means that, at $t = 0$, the system experiences a δ -function spike in the shear rate

$$\frac{\partial v_x}{\partial y} = -\epsilon \delta(t). \quad (\text{F.3.6})$$

We can compute the time-dependent response of the shear stress, $\sigma_{xy}(t)$, to the sudden change from $\mathcal{H}_1$ to $\mathcal{H}_0$:

$$\langle \sigma_{xy}(t) \rangle = -\epsilon \frac{1}{V k_B T} \langle \sigma_{xy}(0) \sigma_{xy}(t) \rangle. \quad (\text{F.3.7})$$

By combining Eqs. (F.3.6) and (F.3.7) with Eq. (2.5.14), we immediately see that the steady-state stress, σ_{xy} , that results from a steady shear is given by

$$\sigma_{xy} = \frac{\partial v_x}{\partial y} \times \frac{1}{V k_B T} \int_0^\infty dt \langle \sigma_{xy}(0) \sigma_{xy}(t) \rangle, \quad (\text{F.3.8})$$

and the resulting expression for the shear viscosity η is

$$\eta = \frac{1}{V k_B T} \int_0^\infty dt \langle \sigma_{xy}(0) \sigma_{xy}(t) \rangle. \quad (\text{F.3.9})$$

F.4 Elastic constants

A liquid flows under the influence of shear forces. A solid does not. Rather, any small deformation of a solid induces an elastic response (stress) that counteracts it. This elastic stress is proportional to the applied deformation (strain). The constants of proportionality between stress and strain (to be defined more precisely below) are called the elastic constants. Below we discuss how to measure these constants by computer simulation. For the sake of simplicity, we limit the discussion to crystals on isotropic (hydrostatic) pressure.

When considering the effect of the strain on the free energy of a solid, it is essential to introduce the so-called *Lagrangian strain tensor* (see, e.g., [182]).¹

¹ Confronted with the finiteness of the Greek alphabet, we use the symbol η to denote the Lagrangian strain tensor. This symbol can easily be confused with the scalar viscosity η .

The reason is that, on a local scale, all changes in free energy are due to changes in the *distances* between the particles that make up the solid. And the quantity that measures this change is precisely the Lagrangian strain. We start with the relation between new and old coordinates due to an elastic deformation:

$$\mathbf{r}' = (1 + \boldsymbol{\epsilon})\mathbf{r}, \quad (\text{F.4.1})$$

where

$$\epsilon_{\alpha\beta} \equiv \frac{\partial u_\alpha}{\partial x_\beta} \quad (\text{F.4.2})$$

is the (conventional) strain tensor. It measures the variation of the displacement field $\mathbf{u}$ with the original coordinate $\mathbf{r}$. Due to the strain, the distance r_{ij} separating two points i and j in the solid is changed. The new squared distance is then related to the old distance by

$$\begin{aligned} r_{ij}'^2 &= \mathbf{r}_{ij} \left(1 + \boldsymbol{\epsilon}^T \right) (1 + \boldsymbol{\epsilon}) \mathbf{r}_{ij} \\ &= \mathbf{r}_{ij} \left(1 + \boldsymbol{\epsilon}^T + \boldsymbol{\epsilon} + \boldsymbol{\epsilon}^T \boldsymbol{\epsilon} \right) \mathbf{r}_{ij} \\ &\equiv \mathbf{r}_{ij} (1 + 2\boldsymbol{\eta}) \mathbf{r}_{ij}. \end{aligned}$$

This defines the Lagrangian strain tensor $\boldsymbol{\eta}$. The new volume V' of the system is related to the original volume V_0 by

$$V' = V_0 \det(1 + \boldsymbol{\epsilon}) \quad (\text{F.4.3})$$

or

$$V' = V_0 \sqrt{\det(1 + 2\boldsymbol{\eta})}. \quad (\text{F.4.4})$$

We now expand the Helmholtz free-energy (F) per unit of (undeformed) volume (V) in powers of the Lagrangian strain parameters $\boldsymbol{\eta}$:

$$\begin{aligned} F(\boldsymbol{\eta})/V &= V^{-1} \left[F(0) + \frac{\partial F}{\partial \eta_{\alpha\beta}} \eta_{\alpha\beta} + \frac{1}{2} \frac{\partial^2 F}{\partial \eta_{\alpha\beta} \partial \eta_{\gamma\delta}} \eta_{\alpha\beta} \eta_{\gamma\delta} + \cdots \right] \\ &= V^{-1} F(0) + C_{\alpha\beta}^{(1)} \eta_{\alpha\beta} + \frac{1}{2} C_{\alpha\beta\gamma\delta}^{(2)} \eta_{\alpha\beta} \eta_{\gamma\delta} + \cdots. \end{aligned} \quad (\text{F.4.5})$$

This equation defines the (second-order) elastic constants $C_{\alpha\beta\gamma\delta}^{(2)}$. To compute the elastic constants numerically, we need a *microscopic* expression for the η -dependence of F . To derive such a relation, we must consider in detail what a deformation of the system does to the partition function. Let us first consider the deformed system. The partition function of this system (ignoring constants, such as h^{-3N}) is equal to

$$Q(\boldsymbol{\eta}) = \int d\mathbf{p}^N d\mathbf{r}^N \exp \left[-\beta \mathcal{H}(\mathbf{p}^N, \mathbf{r}^N) \right]. \quad (\text{F.4.6})$$

This partition function depends on the deformation through the boundary conditions of the integral over the coordinates. This is not very convenient when computing derivatives with respect to the strain. Therefore, we first express the partition function of the deformed system in terms of coordinates and momenta of the original, undeformed system. We can express the coordinates ($\mathbf{r}_i$) and velocities ($\dot{\mathbf{r}}_i$) in this system in terms of the strain tensor $\mathbf{h} \equiv (1 + \boldsymbol{\epsilon})$, and the original coordinates ($\mathbf{r}_{0,i}$) and velocities ($\dot{\mathbf{r}}_{0,i}$):

$$\begin{aligned}\mathbf{r}_i &= \mathbf{h}\mathbf{r}_{0,i} \\ \dot{\mathbf{r}}_i &= \mathbf{h}\dot{\mathbf{r}}_{0,i}\end{aligned}\tag{F.4.7}$$

The kinetic energy, $\mathcal{K} = \sum \frac{1}{2}m_i\dot{\mathbf{r}}_i^2$, can be written as

$$\begin{aligned}\mathcal{K} &= \sum \frac{1}{2}m_i\dot{\mathbf{r}}_i^2 \\ &= \sum \frac{1}{2}m_i\dot{\mathbf{r}}_{0,i}(\mathbf{h}^T\mathbf{h})\dot{\mathbf{r}}_{0,i} \equiv \sum \frac{1}{2}m_i\dot{\mathbf{r}}_{0,i} \cdot \mathbf{G} \cdot \dot{\mathbf{r}}_{0,i},\end{aligned}\tag{F.4.8}$$

where $\mathbf{h}^T = (1 + \boldsymbol{\epsilon}^T)$ is the transverse of $\mathbf{h}$ and $\mathbf{G} = \mathbf{h}^T\mathbf{h}$ is the metric tensor. From the definition of $\mathbf{h}$ it follows that $\mathbf{G} = (1 + 2\boldsymbol{\eta})$. We can now write down the generalized momentum $\mathbf{p}_{0,i}$ conjugate to the coordinate $\mathbf{r}_{0,i}$ (see Appendix A):

$$\mathbf{p}_{0,i}^\alpha = \left(\frac{\partial \mathcal{K}}{\partial \dot{\mathbf{r}}_{0,i}^\alpha} \right) = m_i G_{\alpha\beta} \dot{\mathbf{r}}_{0,i}^\beta\tag{F.4.9}$$

and hence

$$\begin{aligned}\mathcal{K} &= \sum \frac{1}{2}m_i\dot{\mathbf{r}}_{0,i} \cdot \mathbf{G} \cdot \dot{\mathbf{r}}_{0,i} \\ &= \sum \frac{1}{2m_i}\mathbf{p}_{0,i} \cdot \mathbf{G}^{-1} \cdot \mathbf{p}_{0,i} \\ &= \sum \frac{1}{2m_i}\mathbf{p}_{0,i} \cdot (1 + 2\boldsymbol{\eta})^{-1} \cdot \mathbf{p}_{0,i}.\end{aligned}\tag{F.4.10}$$

As

$$\begin{aligned}\mathbf{p}_i &= m_i\dot{\mathbf{r}}_i = m_i\mathbf{h}\dot{\mathbf{r}}_{0,i} = (\mathbf{h}^T)^{-1}\mathbf{p}_{0,i} \\ \mathbf{r}_i &= \mathbf{h}\mathbf{r}_{0,i},\end{aligned}\tag{F.4.11}$$

the Jacobian of the transformation between $\{\mathbf{p}^N, \mathbf{r}^N\}$ and $\{\mathbf{p}_0^N, \mathbf{r}_0^N\}$ is equal to 1. Hence, we can write

$$\begin{aligned}\mathcal{Q}(\eta) &= \int d\mathbf{p}^N d\mathbf{r}^N \exp \left[-\beta \mathcal{H}(\mathbf{p}^N, \mathbf{r}^N) \right] \\ &= \int d\mathbf{p}_0^N d\mathbf{r}_0^N \exp \left\{ -\beta \left[\sum \frac{1}{2m_i} \mathbf{p}_{0,i} \cdot (1 + 2\boldsymbol{\eta})^{-1} \cdot \mathbf{p}_{0,i} + \mathcal{U}(\mathbf{r}_0^N; \eta) \right] \right\}.\end{aligned}\tag{F.4.12}$$

Now the dependence of $Q(\boldsymbol{\eta})$ on $\boldsymbol{\eta}$ is only contained in the Hamiltonian. We can now explicitly carry out the differentiation with respect to $\boldsymbol{\eta}$. Using

$$\begin{aligned} \left(\frac{\partial U}{\partial \eta_{\alpha\beta}} \right) &= \sum_{i < j} \left(\frac{\partial U}{\partial r_{ij}^2} \right) \left(\frac{\partial r_{ij}^2}{\partial \eta_{\alpha\beta}} \right) = \sum_{i < j} \left(\frac{\partial U}{\partial r_{ij}} \right) \frac{r_{0,ij}^\alpha r_{0,ij}^\beta}{r_{ij}} \\ &= \left(\mathbf{h}^{-1} \sum_{i < j} \left(\frac{\partial U}{\partial r_{ij}} \right) \frac{\mathbf{r}_{ij} \mathbf{r}_{ij}}{r_{ij}} (\mathbf{h}^T)^{-1} \right)_{\alpha\beta} \end{aligned} \quad (\text{F.4.13})$$

and

$$\begin{aligned} \sum \frac{1}{2m_i} \mathbf{p}_{0,i} \cdot \left(\frac{\partial G^{-1}}{\partial \eta_{\alpha\beta}} \right) \cdot \mathbf{p}_{0,i} &= - \sum \frac{1}{m_i} \left(\mathbf{p}_{0,i} \cdot \mathbf{G}^{-1} \right)_\alpha \left(\mathbf{G}^{-1} \cdot \mathbf{p}_{0,i} \right)_\beta \\ &= - \left(\mathbf{h}^{-1} \left(\sum \frac{1}{m_i} \mathbf{p} \mathbf{p} \right) (\mathbf{h}^T)^{-1} \right)_{\alpha\beta}, \end{aligned} \quad (\text{F.4.14})$$

we obtain

$$\left(\frac{\partial F}{\partial \eta_{\alpha\beta}} \right) = - \left(\mathbf{h}^{-1} \left\langle \sum \frac{1}{m_i} \mathbf{p} \mathbf{p} + \sum_{j < i} \mathbf{r}_{ij} \mathbf{f}_{ij} \right\rangle (\mathbf{h}^T)^{-1} \right)_{\alpha\beta}. \quad (\text{F.4.15})$$

From this, it follows immediately that

$$\begin{aligned} C_{\alpha\beta}^{(1)} &\equiv \left(\frac{\partial F}{\partial \eta_{\alpha\beta}} \right) \\ &= \frac{V'}{V} \left[(1 + \boldsymbol{\epsilon})^{-1} \boldsymbol{\sigma} (1 + \boldsymbol{\epsilon}^T)^{-1} \right]_{\alpha\beta} \\ &= \sqrt{\det(1 + 2\boldsymbol{\eta})} \left[(1 + \boldsymbol{\epsilon})^{-1} \boldsymbol{\sigma} (1 + \boldsymbol{\epsilon}^T)^{-1} \right]_{\alpha\beta}, \end{aligned} \quad (\text{F.4.16})$$

where

$$\sigma_{\gamma\delta} = - \frac{1}{V'} \sum_i \left[\frac{p_{i\gamma} p_{i\delta}}{m_i} + \sum_{j < i} r_{ij\gamma} f_{ij\delta} \right] \quad (\text{F.4.17})$$

denotes the microscopic stress in the deformed system. Note that $\boldsymbol{\sigma}$ can be measured in a simulation, while $\boldsymbol{\epsilon}$ is fixed by the applied strain. For an undeformed system $C^{(1)}$ is simply equal to $-P$, where P is the hydrostatic pressure. From Eq. (F.4.16), it also follows that the constant of proportionality between the stress σ_{ij} and the linear strain ϵ_{rs} is given by

$$\left(\frac{\partial \sigma_{\alpha\beta}}{\partial \epsilon_{\gamma\delta}} \right) = (\sigma_{\alpha\delta} \delta_{\beta\gamma} + \sigma_{\beta\delta} \delta_{\alpha\gamma} - \sigma_{\alpha\beta} \delta_{\gamma\delta}) + C_{\alpha\beta\gamma\delta}^{(2)}. \quad (\text{F.4.18})$$

To determine the second-order elastic constants, $C_{\alpha\beta\gamma\delta}^{(2)}$, we must determine the initial linear dependence of $C_{\alpha\beta}^{(1)}$ on $\eta_{\gamma\delta}$. This technique for measuring the elastic constants is simple and also quite accurate (see, e.g., [720]). However, several computations are needed to measure the different elastic constants. The lower the symmetry of the crystal, the more calculations are needed. This can be avoided by directly considering the microscopic expression for the $C_{\alpha\beta\gamma\delta}^{(2)}$. Such an expression was derived by Squire et al. [721]:

$$C_{\alpha\beta\gamma\delta}^{(2)} = -\frac{1}{Vk_B T} \langle \Delta\sigma_{\alpha\beta} \Delta\sigma_{\gamma\delta} \rangle + 2\rho k_B T (\delta_{\alpha\gamma} \delta_{\beta\delta} + \delta_{\alpha\delta} \delta_{\beta\gamma}) + 4 \sum_{i < j, k < l} \left\langle \frac{\partial^2 U}{\partial r_{ij}^2 \partial r_{kl}^2} r_{ij\alpha} r_{ij\beta} r_{kl\gamma} r_{kl\delta} \right\rangle. \quad (\text{F.4.19})$$

Using Eq. (F.4.19), it takes only a single simulation to measure *all* elastic constants. Unfortunately, the statistical errors in evaluating this fluctuation expression are usually larger than those obtained when computing Eq. (F.4.16). The problems with statistics appear even worse in constant-stress MD simulations on the same system, where the elastic compliances (rather than the moduli) are determined from fluctuations in the box shape [722]. Eq. (F.4.19) can only be used if the intermolecular potentials are everywhere continuous. However, a fluctuation expression that works for hard-core systems has been developed by Farago and Kantor [723]. More details about the numerical evaluation of elastic constants can be found in [721], [436], [723], and [724], while the general framework of elastic properties at finite temperature is discussed in [182].